

ACR21347

## Antimony(III) oxide

### SECTION 1. IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

|                               |                                                                                                                                                                                                                                                                    |
|-------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 产品说明:<br>Product Description: | 三氧化二锑<br>Antimony(III) oxide                                                                                                                                                                                                                                       |
| Cat No. :                     | 213470000; 213470010; 213470050; 213471000; 213470051                                                                                                                                                                                                              |
| Synonyms                      | Antimony trioxide                                                                                                                                                                                                                                                  |
| CAS No                        | 1309-64-4                                                                                                                                                                                                                                                          |
| Molecular Formula             | O3 Sb2                                                                                                                                                                                                                                                             |
| Supplier                      | <b>UK entity/business name</b><br>Fisher Scientific UK<br>Bishop Meadow Road,<br>Loughborough, Leicestershire LE11 5RG, United Kingdom<br><br><b>EU entity/business name</b><br>Thermo Fisher Scientific<br>Janssen Pharmaceuticaaan 3a, 2440 Geel, Belgium        |
| Emergency Telephone Number    | For information <b>US</b> call: 001-800-227-6701 / <b>Europe</b> call: +32 14 57 52 11<br>Emergency Number <b>US</b> :001-201-796-7100 / <b>Europe</b> : +32 14 57 52 99<br><b>CHEMTREC</b> Tel. No. <b>US</b> :001-800-424-9300 / <b>Europe</b> :001-703-527-3887 |
| E-mail address                | begel.sdsdesk@thermofisher.com                                                                                                                                                                                                                                     |
| Recommended Use               | Laboratory chemicals.                                                                                                                                                                                                                                              |
| Uses advised against          | No Information available                                                                                                                                                                                                                                           |

### SECTION 2. HAZARD IDENTIFICATION

|                                                           |                     |                  |
|-----------------------------------------------------------|---------------------|------------------|
| Physical State<br>Powder Solid                            | Appearance<br>White | Odor<br>Odorless |
| <b>Emergency Overview</b><br>Suspected of causing cancer. |                     |                  |

#### Classification of the substance or mixture

|                 |            |
|-----------------|------------|
| Carcinogenicity | Category 2 |
|-----------------|------------|

#### Label Elements



Signal Word

Warning

Hazard Statements

H351 - Suspected of causing cancer

**Precautionary Statements****Prevention**

P201 - Obtain special instructions before use

P202 - Do not handle until all safety precautions have been read and understood

P280 - Wear protective gloves/protective clothing/eye protection/face protection

**Response**

P308 + P313 - IF exposed or concerned: Get medical advice/attention

**Storage**

P403 - Store in a well-ventilated place

**Disposal**

P501 - Dispose of contents/ container to an approved waste disposal plant

**Physical and Chemical Hazards**

None identified.

**Health Hazards**

Suspected of causing cancer.

**Environmental hazards**

Contains no substances known to be hazardous to the environment or not degradable in waste water treatment plants. Will likely be mobile in the environment due to its water solubility. Is not likely mobile in the environment due its low water solubility. The product is water soluble, and may spread in water systems. Spillage unlikely to penetrate soil.

**Other Hazards**

This product does not contain any known or suspected endocrine disruptors.

**SECTION 3. COMPOSITION/INFORMATION ON INGREDIENTS**

| Component         | CAS No    | Weight % |
|-------------------|-----------|----------|
| Antimony trioxide | 1309-64-4 | >95      |
| Lead monoxide     | 1317-36-8 | <0.1     |
| Arsenic trioxide  | 1327-53-3 | <0.1     |

**Note**

Note 1: The concentration stated or, in the absence of such concentrations, the generic concentrations of this Regulation (Table 3.1) or the generic concentrations of Directive 1999/45/EC (Table 3.2), are the percentages by weight of the metallic element calculated with reference to the total weight of the mixture

**SECTION 4. FIRST AID MEASURES****General Advice**

If symptoms persist, call a physician.

**Eye Contact**

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.

**Skin Contact**

Wash off immediately with plenty of water for at least 15 minutes. If skin irritation persists, call a physician.

**Inhalation**

Remove to fresh air. If not breathing, give artificial respiration. Get medical attention if symptoms occur.

**Ingestion**

Clean mouth with water and drink afterwards plenty of water. Get medical attention if symptoms occur.

**Most important symptoms and effects**

None reasonably foreseeable.

**Self-Protection of the First Aider**

Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination.

**Notes to Physician**

Treat symptomatically.

**SECTION 5. FIRE-FIGHTING MEASURES****Suitable Extinguishing Media**

Water spray, carbon dioxide (CO<sub>2</sub>), dry chemical, alcohol-resistant foam.

**Extinguishing media which must not be used for safety reasons**

No information available.

**Specific Hazards Arising from the Chemical**

Thermal decomposition can lead to release of irritating gases and vapors. Keep product and empty container away from heat and sources of ignition.

**Protective Equipment and Precautions for Firefighters**

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

**SECTION 6. ACCIDENTAL RELEASE MEASURES****Personal Precautions**

Ensure adequate ventilation. Use personal protective equipment as required. Avoid dust formation.

**Environmental Precautions**

Do not flush into surface water or sanitary sewer system. Should not be released into the environment. Do not allow material to contaminate ground water system.

**Methods for Containment and Clean Up**

Sweep up and shovel into suitable containers for disposal. Keep in suitable, closed containers for disposal.

Refer to protective measures listed in Sections 8 and 13.

**SECTION 7. HANDLING AND STORAGE****Handling**

Wear personal protective equipment/face protection. Ensure adequate ventilation. Avoid dust formation. Do not get in eyes, on skin, or on clothing. Avoid ingestion and inhalation.

**Storage**

Keep containers tightly closed in a dry, cool and well-ventilated place.

**Specific Use(s)**

Use in laboratories

**SECTION 8. EXPOSURE CONTROLS/PERSONAL PROTECTION****Control Parameters**

| Component         | China | Taiwan                      | Thailand                    | Hong Kong |
|-------------------|-------|-----------------------------|-----------------------------|-----------|
| Antimony trioxide | -     | TWA: 0.5 mg/m <sup>3</sup>  | TWA: 0.5 mg/m <sup>3</sup>  | -         |
| Lead monoxide     | -     | TWA: 0.05 mg/m <sup>3</sup> | TWA: 0.05 mg/m <sup>3</sup> | -         |
| Arsenic trioxide  | -     | TWA: 0.01 mg/m <sup>3</sup> | TWA: 0.01 mg/m <sup>3</sup> | -         |

**SAFETY DATA SHEET****Antimony(III) oxide**

| Component         | ACGIH TLV                                                 | OSHA PEL                             | NIOSH                                                         | The United Kingdom                                                                           | European Union |
|-------------------|-----------------------------------------------------------|--------------------------------------|---------------------------------------------------------------|----------------------------------------------------------------------------------------------|----------------|
| Antimony trioxide | TWA: 0.02 mg/m <sup>3</sup><br>TWA: 0.5 mg/m <sup>3</sup> | (Vacated) TWA: 0.5 mg/m <sup>3</sup> | IDLH: 50 mg/m <sup>3</sup><br>TWA: 0.5 mg/m <sup>3</sup>      | STEL: 1.5 mg/m <sup>3</sup> 15 min<br>TWA: 0.5 mg/m <sup>3</sup> 8 hr                        |                |
| Lead monoxide     | TWA: 0.05 mg/m <sup>3</sup>                               |                                      | IDLH: 100 mg/m <sup>3</sup><br>TWA: 0.050 mg/m <sup>3</sup>   | STEL: 0.45 mg/m <sup>3</sup> 15 min<br>TWA: 0.15 mg/m <sup>3</sup> 8 hr                      |                |
| Arsenic trioxide  | TWA: 0.01 mg/m <sup>3</sup>                               |                                      | IDLH: 5 mg/m <sup>3</sup><br>Ceiling: 0.002 mg/m <sup>3</sup> | STEL: 0.3 mg/m <sup>3</sup> 15 min<br>TWA: 0.1 mg/m <sup>3</sup> 8 hr<br>Carc. except Arsine |                |

Legend

ACGIH - American Conference of Governmental Industrial Hygienists

OSHA - Occupational Safety and Health Administration

NIOSH: NIOSH - National Institute for Occupational Safety and Health

**Monitoring methods**

BS EN 14042:2003 Title Identifier: Workplace atmospheres. Guide for the application and use of procedures for the assessment of exposure to chemical and biological agents. MDHS14/3 General methods for sampling and gravimetric analysis of respirable and inhalable dust MDHS 99 Metals in air by ICP-AES MDHS 91 Metals and metalloids in workplace air by X-ray fluorescence spectrometry

**Exposure Controls****Engineering Measures**

Ensure adequate ventilation, especially in confined areas. Ensure that eyewash stations and safety showers are close to the workstation location. Use only under a chemical fume hood. Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source.

**Personal protective equipment**

**Eye Protection** Wear safety glasses with side shields (or goggles) (European standard - EN 166)

**Hand Protection** Protective gloves

| Glove material | Breakthrough time                 | Glove thickness | EU standard | Glove comments        |
|----------------|-----------------------------------|-----------------|-------------|-----------------------|
| Neoprene       | See manufacturers recommendations | -               | EN 374      | (minimum requirement) |

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatibility, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

**Skin and body protection** Wear appropriate protective gloves and clothing to prevent skin exposure

**Respiratory Protection** When workers are facing concentrations above the exposure limit they must use appropriate certified respirators. To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

**Large scale/emergency use** Use a NIOSH/MSHA or European Standard EN 136 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced  
**Recommended Filter type:** Particulates filter conforming to EN 143

**Small scale/Laboratory use** Use a NIOSH/MSHA or European Standard EN 149:2001 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.  
**Recommended half mask:-** Valve filtering: EN405; or; Half mask: EN140; plus filter, EN 141  
When RPE is used a face piece Fit Test should be conducted

|                                 |                                                                                                                                                                                   |
|---------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Hygiene Measures                | Handle in accordance with good industrial hygiene and safety practice.                                                                                                            |
| Environmental exposure controls | Prevent product from entering drains. Do not allow material to contaminate ground water system. Local authorities should be advised if significant spillages cannot be contained. |

**SECTION 9. PHYSICAL AND CHEMICAL PROPERTIES**

|                                         |                          |                                   |
|-----------------------------------------|--------------------------|-----------------------------------|
| Appearance                              | White                    |                                   |
| Physical State                          | Powder Solid             |                                   |
| Odor                                    | Odorless                 |                                   |
| Odor Threshold                          | No data available        |                                   |
| pH                                      | No information available |                                   |
| Melting Point/Range                     | 656 °C / 1212.8 °F       |                                   |
| Softening Point                         | No data available        |                                   |
| Boiling Point/Range                     | 1550 °C / 2822 °F        | @ 760 mmHg                        |
| Flash Point                             | No information available | Method - No information available |
| Evaporation Rate                        | Not applicable           | Solid                             |
| Flammability (solid,gas)                | No information available |                                   |
| Explosion Limits                        | No data available        |                                   |
| Vapor Pressure                          | 1.3 hPa @ 574 °C         |                                   |
| Vapor Density                           | Not applicable           | Solid                             |
| Specific Gravity / Density              | No data available        |                                   |
| Bulk Density                            | No data available        |                                   |
| Water Solubility                        | Insoluble in water       |                                   |
| Solubility in other solvents            | No information available |                                   |
| Partition Coefficient (n-octanol/water) |                          |                                   |
| Component                               | log Pow                  |                                   |
| Arsenic trioxide                        | 18.1                     |                                   |
| Autoignition Temperature                | No data available        |                                   |
| Decomposition Temperature               | No data available        |                                   |
| Viscosity                               | Not applicable           | Solid                             |
| Explosive Properties                    | No information available |                                   |
| Oxidizing Properties                    | No information available |                                   |
| Molecular Formula                       | O3 Sb2                   |                                   |
| Molecular Weight                        | 291.42                   |                                   |

**SECTION 10. STABILITY AND REACTIVITY**

|                                  |                                                                      |
|----------------------------------|----------------------------------------------------------------------|
| Stability                        | Stable under normal conditions.                                      |
| Hazardous Reactions              | None under normal processing.                                        |
| Hazardous Polymerization         | Hazardous polymerization does not occur.                             |
| Conditions to Avoid              | Avoid dust formation. Incompatible products. Excess heat.            |
| Materials to avoid               | Strong acids. Strong bases. Reducing Agent. Strong oxidizing agents. |
| Hazardous Decomposition Products | Antimony oxide.                                                      |

**SECTION 11. TOXICOLOGICAL INFORMATION****Product Information**

**SAFETY DATA SHEET****Antimony(III) oxide****(a) acute toxicity;**

| Component         | LD50 Oral                  | LD50 Dermal                  | LC50 Inhalation              |
|-------------------|----------------------------|------------------------------|------------------------------|
| Antimony trioxide | LD50 > 34600 mg/kg ( Rat ) | LD50 > 2000 mg/kg ( Rabbit ) | LC50 > 5.2 mg/L ( Rat ) 4 h  |
| Lead monoxide     | LD50 > 10000 mg/kg ( Rat ) | LD50 > 2000 mg/kg ( Rat )    | LC50 > 5.05 mg/L ( Rat ) 4 h |
| Arsenic trioxide  | LD50 = 20 mg/kg ( Rat )    |                              |                              |

**(b) skin corrosion/irritation;** No data available**(c) serious eye damage/irritation;** No data available**(d) respiratory or skin sensitization;****Respiratory** No data available**Skin** No data available**(e) germ cell mutagenicity;** No data available**(f) carcinogenicity;** Category 2

The table below indicates whether each agency has listed any ingredient as a carcinogen

| Component         | EU           | UK | Germany | IARC     |
|-------------------|--------------|----|---------|----------|
| Antimony trioxide |              |    |         | Group 2B |
| Lead monoxide     |              |    |         | Group 2A |
| Arsenic trioxide  | Carc Cat. 1A |    | Cat. 1  | Group 1  |

**(g) reproductive toxicity;** No data available**(h) STOT-single exposure;** No data available**(i) STOT-repeated exposure;** No data available**Target Organs** None known.**(j) aspiration hazard;** Not applicable  
Solid**Symptoms / effects, both acute and delayed** No information available**SECTION 12. ECOLOGICAL INFORMATION****Ecotoxicity effects**

Contains a substance which is: Very toxic to aquatic organisms. The product contains following substances which are hazardous for the environment. May cause long-term adverse effects in the environment. Do not allow material to contaminate ground water system.

| Component         | Freshwater Fish                             | Water Flea                                                                                              | Freshwater Algae                                                                                                                 | Microtox            |
|-------------------|---------------------------------------------|---------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|---------------------|
| Antimony trioxide | LC50 > 1000 mg/L/96h<br>(Brachydanio rerio) | EC50: 361.5 - 496.0<br>mg/L, 48h Static<br>(Daphnia magna)<br>EC50: > 1000 mg/L, 48h<br>(Daphnia magna) | EC50: 0.65 - 0.81 mg/L,<br>96h<br>(Pseudokirchneriella<br>subcapitata)<br>EC50: 0.63 - 0.8 mg/L,<br>72h<br>(Pseudokirchneriella) | EC50 > 3.5 mg/L 7 h |

**SAFETY DATA SHEET****Antimony(III) oxide**

|                  |                                                                                                                                                                                     |                                                                        |              |                                                                                                                         |
|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------|--------------|-------------------------------------------------------------------------------------------------------------------------|
|                  |                                                                                                                                                                                     |                                                                        | subcapitata) |                                                                                                                         |
| Lead monoxide    | Pimephales promelas:<br>LC50=0.3 mg/L 96h                                                                                                                                           | EC50=0.13 mg/L 48h                                                     |              |                                                                                                                         |
| Arsenic trioxide | LC50: = 135 mg/L, 96h<br>(Pimephales promelas)<br>LC50: > 1000 mg/L, 96h<br>static (Oncorhynchus<br>mykiss)<br>LC50: 18.8 - 21.4 mg/L,<br>96h flow-through<br>(Oncorhynchus mykiss) | EC50 = 0.038 mg/L 24h<br>EC50 = 0.96 mg/L 96h<br>EC50 = 0.038 mg/L 24h |              | EC50 = 31.43 mg/L 60<br>min<br>EC50 = 33.39 mg/L 30<br>min<br>EC50 = 43.56 mg/L 15<br>min<br>EC50 = 73.73 mg/L 5<br>min |

**Persistence and Degradability****Persistence  
Degradation in sewage  
treatment plant**

Product contains heavy metals. Discharge into the environment must be avoided. Special pre-treatment is necessary based on information available, May persist, Insoluble in water.  
Contains substances known to be hazardous to the environment or not degradable in waste water treatment plants.

**Bioaccumulative Potential**

May have some potential to bioaccumulate; Product has a high potential to bioconcentrate

| Component        | log Pow | Bioconcentration factor (BCF) |
|------------------|---------|-------------------------------|
| Arsenic trioxide | 18.1    | 80 - 236 dimensionless        |

**Mobility in soil**

The product is water soluble, and may spread in water systems Spillage unlikely to penetrate soil Will likely be mobile in the environment due to its water solubility Is not likely mobile in the environment due its low water solubility Highly mobile in soils

**Endocrine Disruptor Information  
Persistent Organic Pollutant  
Ozone Depletion Potential**

This product does not contain any known or suspected endocrine disruptors  
This product does not contain any known or suspected substance  
This product does not contain any known or suspected substance

**SECTION 13. DISPOSAL CONSIDERATIONS****Waste from Residues/Unused  
Products**

Waste is classified as hazardous. Dispose of in accordance with the European Directives on waste and hazardous waste. Dispose of in accordance with local regulations.

**Contaminated Packaging**

Dispose of this container to hazardous or special waste collection point.

**Other Information**

Do not flush to sewer. Waste codes should be assigned by the user based on the application for which the product was used. Do not empty into drains.

**SECTION 14. TRANSPORT INFORMATION****Road and Rail Transport**

Not Regulated

**IMDG/IMO**

Not regulated

**IATA**

Not regulated

**Special Precautions for User**

No special precautions required

**SECTION 15. REGULATORY INFORMATION**

## Antimony(III) oxide

## International Inventories

X = listed, China (IECSC), Europe (EINECS/ELINCS/NLP), U.S.A. (TSCA), Canada (DSL/NDSL), Philippines (PICCS), Japan (ENCS), Japan (ISHL), Australia (AICS), Korea (KECL).

| Component         | The Inventory of Hazardous Chemicals (2015 Edition) | List of dangerous goods GB 12268 - 2012 | TCSI | IECSC | EINECS    | TSCA | DSL | PICCS | ENCS | ISHL | AICS | KECL     |
|-------------------|-----------------------------------------------------|-----------------------------------------|------|-------|-----------|------|-----|-------|------|------|------|----------|
| Antimony trioxide | -                                                   | X                                       | X    | X     | 215-175-0 | X    | X   | X     | X    | X    | X    | KE-09846 |
| Lead monoxide     | X                                                   | -                                       | X    | X     | 215-267-0 | X    | X   | X     | X    | X    | X    | KE-21926 |
| Arsenic trioxide  | X                                                   | X                                       | X    | X     | 215-481-4 | X    | X   | X     | X    | X    | X    | KE-09858 |

## Note

Note 1: The concentration stated or, in the absence of such concentrations, the generic concentrations of this Regulation (Table 3.1) or the generic concentrations of Directive 1999/45/EC (Table 3.2), are the percentages by weight of the metallic element calculated with reference to the total weight of the mixture

| Component        | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements |
|------------------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| Arsenic trioxide |                                                                                           | 0.1 tonne                                                                                |

## National Regulations

| Component                            | Toxic Chemical Substances Control Act                                   |
|--------------------------------------|-------------------------------------------------------------------------|
| Arsenic trioxide<br>1327-53-3 (<0.1) | Class I (1 wt%)<br>Class II (1 wt%)<br>Class III (1 wt%)<br>TRQ = 50 kg |

## SECTION 16. OTHER INFORMATION

**Creation Date** 22-Dec-2009  
**Revision Date** 07-Apr-2024  
**Revision Summary** SDS sections updated.

## Training Advice

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

Use of personal protective equipment, covering appropriate selection, compatibility, breakthrough thresholds, care, maintenance, fit and standards.

First aid for chemical exposure, including the use of eye wash and safety showers.

Chemical incident response training.

## Legend

**CAS** - Chemical Abstracts Service

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**IECSC** - Chinese Inventory of Existing Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDSL** - Canadian Domestic Substances List/Non-Domestic Substances List

**ENCS** - Japanese Existing and New Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

**NZIoC** - New Zealand Inventory of Chemicals

**WEL** - Workplace Exposure Limit

**ACGIH** - American Conference of Governmental Industrial Hygienists

**DNEL** - Derived No Effect Level

**RPE** - Respiratory Protective Equipment

**TWA** - Time Weighted Average

**IARC** - International Agency for Research on Cancer

**PNEC** - Predicted No Effect Concentration

**LD50** - Lethal Dose 50%



## Antimony(III) oxide

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**LC50** - Lethal Concentration 50%**NOEC** - No Observed Effect Concentration**PBT** - Persistent, Bioaccumulative, Toxic**EC50** - Effective Concentration 50%**POW** - Partition coefficient Octanol:Water**vPvB** - very Persistent, very Bioaccumulative**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association**ADR** - European Agreement Concerning the International Carriage of Dangerous Goods by Road**OECD** - Organisation for Economic Co-operation and Development**BCF** - Bioconcentration factor**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code**MARPOL** - International Convention for the Prevention of Pollution from Ships**ATE** - Acute Toxicity Estimate**VOC** - (Volatile Organic Compound)**Key literature references and sources for data**<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**